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Trek de derdemachtswortels in $\mathbb{C}$:

$$z^3 = 27 \operatorname{cis} \left(\frac{5\pi}{6} \right)$$

$$\Rightarrow z_k = 3 \operatorname{cis} \left(\frac{\frac{5\pi}{6} + 2k\pi}{3} \right), k = 0, 1, 2$$

$$\Rightarrow z_k = 3 \operatorname{cis} \left(\frac{5\pi}{18} + \frac{2k\pi}{3} \right), k = 0, 1, 2$$

$$\Rightarrow z_0 = 3 \operatorname{cis} \left(\frac{5\pi}{18} \right)$$

$$\Rightarrow z_1 = 3 \operatorname{cis} \left(\frac{17\pi}{18} \right)$$

$$\Rightarrow z_2 = 3 \operatorname{cis} \left(\frac{29\pi}{18} \right)$$

$$\Rightarrow \boxed{z_0 = 3 \operatorname{cis} \left(\frac{5\pi}{18} \right), z_1 = 3 \operatorname{cis} \left(\frac{17\pi}{18} \right), z_2 = 3 \operatorname{cis} \left(\frac{29\pi}{18} \right)}$$

References